

Kyle B.R. Bennett

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PROFESSIONAL SUMMARY

Mechatronics Engineer specializing in embedded C/C++ software development for sensor-driven systems, real-time data acquisition, and hardware-software validation in R&D environments. Hands-on experience with PLC programming, sensor integration, control logic development, and CAD-based design and prototyping. Growing interest in robotics platforms and biometric/embedded control systems.

PROFESSIONAL EXPERIENCE

Head Product Design Engineer & Founder — B3N Industries LLC

Pittsburgh, PA | May 2026–Present

- Founded B3N Industries and now serve as Head Product Design Engineer, leading embedded mechatronic prototype development that integrates microcontroller firmware, sensors, RFID, electromechanical actuation, and CAD-designed enclosures.
- Directed iterative design, fabrication, and bench validation of the F.O.C.U.S. Wallet, managing requirements documentation, test protocols, and design revisions across a 12-person team spanning engineering, software, design, QA, and communications.

Research and Design Intern — Sensus (Xylem)

California, PA | 2025

- Improved metering accuracy by approximately 30% by designing and implementing an embedded C/C++ sensor-driven system integrating firmware with real-time data acquisition.
- Cut bench testing time from 2 hours to 40 minutes (a 67% reduction) by streamlining validation procedures while ensuring compliance with sponsor requirements and engineering best practices.

Research & Development Intern — Rainbarrel Energy

California, PA | 2023–2024

- Improved prototype efficiency by at least 30% by building and bench-testing sensor-based systems focused on energy optimization.
- Directly influenced final prototype design changes by conducting data analysis that uncovered key system inefficiencies.

SELECTED ENGINEERING PROJECTS

F.O.C.U.S. Wallet — Smart Embedded Financial Feedback System (B3N Industries)

- Validated system reliability across 12 test cycles over 6 months of iterative development on a C++-based embedded system integrating RFID tracking with real-time haptic and audio feedback.
- Enabled scalable real-time event handling by architecting modular hardware-software integration between microcontroller firmware and a mobile interface.

B3N Spaces — Self-Hosted 360° Virtual Tour Platform (B3N Industries)

- Eliminated recurring per-tour subscription costs by building a self-hosted 360° virtual tour platform on Pannellum as an alternative to Matterport/Kuula, using a JSON-driven scene system requiring zero code changes to onboard a new property.
- Delivered precise, professional-feeling navigation by implementing per-hotspot arrival orientation (custom targetYaw/targetPitch), then made each tour client-ready by routing it to a unique URL for direct iframe embedding on real estate websites.

EMG-Controlled Robotic Arm — Independent Research Project

- Built a robotic arm controlled by real-time EMG (muscle-activity) sensor input, applying signal processing and embedded control logic to translate biometric signals into motion — presented alongside original research on Da Vinci's contributions to robotics at an undergraduate research conference.

TECHNICAL SKILLS

Languages: C++ (Primary), Embedded C/C++, Python, Java, Kotlin, Pascal

Robotics & Embedded: Robot Operating System (ROS Fundamentals), Arduino, Sensor Integration, Real-Time Data Processing, Embedded Systems, Control Logic Development, Hardware-Software Integration

Controls & Instrumentation: Allen-Bradley Programmable Logic Controller (PLC), Siemens S7 (TIA Portal), LabVIEW, MATLAB

CAD & 3D Design: SolidWorks, AutoCAD, AutoCAD Electrical, Inventor, Fusion 360, KiCad, Revit, Spatial Modeling, CAD-Based Simulation

Tools & Platforms: Git, GitHub, GitLab, Linux, AWS Fundamentals (EC2, S3)

EDUCATION & CERTIFICATIONS

B.S., Mechatronics Engineering Technology (Minor: Graphic Design)

December 2025

Pennsylvania Western University, California, PA

Certified SolidWorks Associate (CSWA), 2025 | OSHA 10-Hour General Industry, 2025 | First Aid & CPR – American Red Cross, 2024